

CINCINNATI LUNKEN, OH, USA

WMO: 724297

Lat: 39.103N

Lon: 84.419W

Elev: 490

StdP: 14.44

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93812

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.1	12.6	-3.1	4.7	9.2	1.9	6.2	14.7	24.2	43.2	21.1	38.1	6.2	290	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	92.4	75.0	89.8	74.4	87.6	73.5	78.3	87.7	77.0	85.8	75.7	83.8	8.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	136.2	82.8	74.3	130.7	81.3	73.2	125.4	80.0	42.0	87.7	40.6	86.0	39.3	83.9	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.3	17.9	15.9	0.3	95.6	8.0	3.3	-5.5	98.0	-10.2	100.0	-14.6	101.9	-20.5	104.3		
			-0.6	81.1	7.6	1.7	-6.1	82.3	-10.6	83.2	-14.9	84.2	-20.4	85.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.0	31.9	35.5	44.0	54.9	64.2	72.4	75.8	74.9	68.1	56.3	44.7	36.4	(6)
(7)		DBStd	17.62	11.83	10.89	10.17	8.51	7.48	5.54	4.85	4.78	6.78	8.31	8.91	9.70	(7)
(8)		HDD50	1959	572	419	244	47	3	0	0	0	31	209	434	(8)	
(9)		HDD65	4776	1025	827	653	317	110	10	0	1	45	292	610	886	(9)
(10)		CDD50	3791	12	13	57	193	443	671	800	771	543	226	49	13	(10)
(11)		CDD65	1135	0	0	2	13	85	232	336	307	138	22	0	0	(11)
(12)		CDH74	10451	0	2	28	172	801	2083	3118	2738	1270	234	5	0	(12)
(13)		CDH80	3791	0	0	2	19	211	738	1247	1064	458	52	0	0	(13)
(14)	Wind	WSAvg	6.0	7.7	7.4	7.4	7.3	5.8	5.1	4.7	4.2	4.2	5.1	6.1	6.6	(14)
(15)	Precipitation	PrecAvg	41.1	2.6	2.4	3.9	3.7	4.4	3.6	4.0	3.9	3.3	2.8	3.3	3.1	(15)
(16)		PrecMax	55.0	7.3	6.5	11.1	6.3	10.2	7.4	7.9	6.6	8.5	8.5	7.0	8.0	(16)
(17)		PrecMin	28.7	0.5	0.3	1.3	0.7	0.6	0.8	0.6	0.7	0.5	0.6	0.6	0.4	(17)
(18)		PrecStd	6.3	1.4	1.5	2.0	1.6	2.4	1.5	2.1	1.7	2.2	1.8	1.6	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.8	69.8	78.2	82.6	88.2	92.4	96.4	95.8	93.8	85.5	73.6	66.3	(19)
(20)		0.4%	MCWB	59.4	59.7	62.7	65.9	72.0	75.5	78.2	74.8	70.8	69.4	62.0	60.3	(20)
(21)		2%	DB	60.8	63.1	72.3	79.0	84.9	89.7	92.3	91.6	89.2	80.0	68.8	60.7	(21)
(22)		2%	MCWB	56.4	54.4	58.9	64.0	69.8	74.5	76.4	74.5	71.9	66.8	59.2	55.6	(22)
(23)		5%	DB	55.7	58.0	66.4	75.2	82.4	87.3	89.5	88.7	85.6	75.7	64.7	56.3	(23)
(24)		5%	MCWB	51.4	51.6	55.8	61.7	69.2	73.2	75.1	74.1	70.5	64.4	57.3	52.1	(24)
(25)		10%	DB	49.7	53.0	61.5	71.3	78.9	84.6	86.9	86.1	81.5	71.5	60.4	51.8	(25)
(26)		10%	MCWB	45.3	46.8	53.6	59.6	67.5	71.9	74.1	72.7	68.9	62.6	53.8	47.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.4	61.8	64.7	69.1	75.0	78.7	80.5	79.4	77.1	73.3	65.2	61.5	(27)
(28)		0.4%	MCDWB	64.1	68.3	75.8	79.5	84.0	88.6	90.7	88.6	85.9	81.8	71.1	65.8	(28)
(29)		2%	WB	57.1	57.0	61.3	66.0	72.8	76.7	78.6	77.8	75.0	69.7	61.6	56.9	(29)
(30)		2%	MCDWB	60.7	61.1	69.2	74.8	81.7	86.1	88.3	86.9	83.4	76.0	66.7	59.7	(30)
(31)		5%	WB	52.0	52.4	58.3	63.7	70.8	75.2	77.3	76.4	73.2	66.8	58.3	52.7	(31)
(32)		5%	MCDWB	55.2	56.9	64.6	71.4	79.3	84.2	86.3	84.6	80.9	73.3	63.5	55.7	(32)
(33)		10%	WB	45.9	46.7	54.3	61.5	68.9	73.8	75.9	75.1	71.3	63.9	54.6	47.8	(33)
(34)		10%	MCDWB	49.1	52.4	60.7	69.4	76.1	82.1	83.9	82.4	78.0	69.3	59.7	51.0	(34)
(35)	Mean Daily Temperature Range		MDBR	15.8	17.8	19.8	22.7	21.2	20.4	19.9	20.7	22.3	22.0	19.8	15.5	(35)
(36)		5% DB	MCDBR	21.0	26.6	27.9	28.2	25.2	23.5	22.4	23.5	27.1	27.7	25.9	21.8	(36)
(37)			MCWBR	17.1	19.5	16.8	15.7	12.5	10.4	8.8	9.2	11.3	14.9	17.7	17.6	(37)
(38)			MCDBR	18.7	23.5	23.9	23.5	21.5	20.5	19.7	19.7	21.7	23.1	22.0	19.5	(38)
(39)		5% WB	MCWBR	16.5	19.4	16.0	14.5	11.5	10.1	8.8	8.9	10.5	13.5	16.8	17.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.329	0.359	0.401	0.445	0.460	0.469	0.464	0.410	0.361	0.332	0.308	(40)	
(41)		taud	2.476	2.388	2.372	2.261	2.176	2.180	2.164	2.162	2.290	2.431	2.472	2.493	(41)	
(42)		Ebn at Noon	274	283	285	278	266	262	258	256	264	268	263	266	(42)	
(43)		Edh at Noon	25	31	35	42	47	47	47	46	38	30	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	572	817	1149	1546	1742	1939	1866	1736	1427	1029	700	486	(44)	
(45)		RadStd	53	104	103	121	153	113	111	94	143	103	81	49	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+211	+130	(47)

Nomenclature: See separate page